



DEPARTMENT: EXECUTIVE MASTER OF PUBLIC HEALTH

Course Number: AEPI 537D

Section Number: 901

Credit Hours: 2

Semester: Spring 2021

Course Title: Introduction to SAS

INSTRUCTOR NAME: Nicole Luisi, MPH MS

INSTRUCTOR CONTACT INFORMATION

Email: nluisi@emory.edu

School Address or

Mailbox Location:

Rollins School of Public Health
Emory University
1518 Clifton Road
Atlanta, Georgia 30322

Office Hours: Weekly TBD (pending class input)

COURSE DESCRIPTION

This course is an introduction to the SAS programming environment. This is an applied computer course that instructs students in the techniques needed to enter data into a database and properly read and process data into a final dataset that is ready for epidemiologic analysis.

This course is required of all Applied Epidemiology students and is offered in the spring semester.

Pre-requisites: BIOS 516D, AEPI 530D or by special permission.

COMPETENCIES AND LEARNING OBJECTIVES

MPH/MSPH Foundational Competencies

This course addresses the following MPH foundational competencies:

- Analyze quantitative data using biostatistics, informatics, computer- based programming and software, as appropriate

Concentration/Track Competencies

This course addresses the following Applied Epidemiology competencies:

- Calculate and interpret basic design-specific measures of association and their standard errors
- Utilize statistical programming packages in preparing scientific reports
- Communicate epidemiologic information in a written scientific report

Learning Objectives

Upon completing this course, students will be able to:

- Input data into SAS from various sources
- Find and deal with inappropriate data values
- Create derived variables
- Merge data from multiple sources
- Perform various data manipulation procedures for desired output
- Determine, and compare between groups, the frequencies of variables

COURSE STRUCTURE

Modules

- Module 1: Intro to SAS, SAS dataset basics, SAS explorer applications, SAS customizations, SAS resources, best practices
- Module 2: Data dataset creation, concatenating and merging datasets
- Module 3: Sub-setting of SAS datasets, data cleaning, creating new variables
- Module 4: Categorical data, performing arithmetic operations, SAS dates and date functions
- Module 5: More on IF/THEN/ELSE, Converting values between character and numeric
- Module 6: Re-naming, labeling, and formatting variables, useful procedures, output delivery system basics
- Module 7: SAS Functions
- Module 8: Using arrays, naming of variables in datasets
- Module 9: Using datasets with multiple observations per subject
- Module 10: More on reporting/output, exporting data
- Module 11: Intro to macros
- Module 12: Intro to graphs/charts/maps, Extra topic(s) TBD

On-campus Meeting Times

There are two synchronous weekends for this class. Attendance is **required** for all weekend class sessions. Both the opening and closing weekends will be virtual. The weekend sessions will be made up of mostly hands-on lab work. Students are expected to participate by attending all sessions from a device with a camera and microphone as well as access to SAS (installed or through a virtual desktop such as Apporto).

- Friday, January 08, 2021 from 4:00 PM–7:00 PM
- Sunday, January 10, 2021 from 8:30 AM–11:30 AM
- Friday, April 09, 2021 from 4:00 PM–7:00 PM
- Sunday, April 11, 2021 from 8:30 AM–11:30 AM

Text/Readings

All course materials will be provided in Canvas via the online course modules.

Optional reference texts you may find useful:

- Delwiche and Slaughter (2012). The little SAS book: A primer (5th edition). Cary, NC: SAS Press. [ISBN-10: 1612903439 or ISBN-13: 978-1-61290-343-9]

Computer Requirements

Computer hardware, software, and peripherals compatible with Canvas usage and Zoom are required. Please note that an iPad or similar mobile device does not provide the needed functionality for all of the Canvas tools.

In addition to the general requirements referenced above, this course requires the use of the following:

- STATISTICAL PROGRAM SAS 9.4 is available to download from Emory Software Express (PC only) or online through Apporto. It may be possible to use another resource such as Citrix, SAS University Edition, or SAS on Demand as long as the requirements for the assignments can be met. Please discuss with the instructor if there are questions about proper access to SAS for this course.

Internet usage is variable and programs like SAS that require significant bandwidth may not run well at certain times of high usage. Therefore, it is very important that you do not wait until the last minute to do your SAS homework, especially if you are using Apporto.

EVALUATION

Grading Overview

Assignments	Points	Due Dates	Notes
Participation in class and online	5%	Ongoing	Must participate during all on campus sessions and complete all assignments (11 SAS labs, 11 quizzes, midterm, article presentation, final exam).
Quizzes	10%	See course schedule	All quizzes must be completed, but lowest score dropped, 10/11 count

			towards grade
SAS Lab Assignments	50%	See course schedule	All labs must be completed, but lowest score dropped, 10/11 count towards grade
Midterm	10%	Opens 2/15 Closes 2/21 at 11:59pm	
Article Presentation	5%	In class 4/9	Article choice due during Week 10
Final Exam	20%	In class 4/11	

Grading Scale

Grade	Meaning	Quality Points
A	93-100	4.0
A-	90-92.9	3.7
B+	85-89.9	3.3
B	80-84.9	3.0
B-	77-79.9	2.7
C	70-76.9	2.0
F	Fail	0.0

ASSIGNMENTS AND ASSESSMENTS

The information below is intended to provide a brief description of the assignments for this course. More detailed information is posted on Canvas.

Data Sets

The data sets used in this course and details about the use of the data sets are located on the Canvas course site.

Optional Reference Materials

A packet of notes will be provided for each lecture/demo. In addition to these notes, optional article links may be provided for reference.

Guidelines and Suggestions

It is recommended that you complete each lab in the real-time with the video and notes, then complete the quiz and lab assignment, referring back to the notes when necessary. You might want to download or print out the assignments before you plan to work on them. This way, if

your internet connection is slow, or if you temporarily cannot reach the course site you will have the assignments in hard copy and can continue to work.

Labs (50% of final grade)

The SAS code submitted as part of every lab assignment must be written such that the instructor is able to generate the student's results by simply changing one "libname" statement. This will be discussed in more detail in the opening on-campus session. SAS code should be well commented to indicate any section and question numbers. SAS code must be copied and pasted from the SAS editor into a Word document and submitted as a Word document (.doc, .docx). All labs must be completed but the lab with the lowest score will be dropped.

Quizzes (10% of final grade)

Each week includes a short quiz based on the weekly lecture/demo materials. All 11 quizzes must be completed. Two attempts will be permitted and the quiz with the lowest score will be dropped.

Midterm Assignment (Due 2/21, 11:59 pm, 10% of final grade)

You will be asked several questions similar to those on the weekly quizzes. You will also be asked to create a SAS program to accomplish several tasks with a prepared dataset. This will be similar to the previous lab assignments but include a range of topics.

Article Presentation (In class 4/9, 5% of final grade)

You will be asked to select a SAS conference proceeding article of your choice from <https://www.lexjansen.com/> and do a brief in-class presentation to walk us through the article on 4/9. This will be discussed in more detail in the opening on-campus session. The presentation should be a big picture overview of the article to introduce the concept. This presentation should be less than 10 minutes. Note that the article of choice must be submitted to the instructor in advance by the end of Week 10.

Final Exam (In class 4/11, 20% of final grade)

You will be asked to create a SAS program to accomplish several tasks with a prepared dataset. This will be similar to the previous lab assignments but include a range of topics.

Participation (5% of final grade)

You must participate in all virtual class sessions. All quizzes, labs, and assignments must be completed on time for full credit – points will be deducted for missing assignments even when the lowest score may be dropped.

ALIGNMENT OF COMPETENCIES AND ASSESSMENTS

MPH/MSPH Foundational Competencies Assessed	Representative Assignment
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Analyze quantitative data using biostatistics, informatics, computer-based programming and software, as appropriate	Final Exam
AEPI Concentration Competencies Assessed	Representative Assignment
Calculate and interpret basic design-specific measures of association and their standard errors	SAS Lab Assignment 3
Utilize statistical programming packages in preparing scientific reports	SAS Lab Assignment 1
Communicate epidemiologic information in a written scientific report	SAS Lab Assignment 9

COURSE POLICIES

Late Assignments

Assignments turned in late **without the instructor's previous knowledge and approval** will be graded down before any corrections are made (10% point reduction after due date/time, additional 10% every 24 hours). If you have an emergency that delays turning in assignments, please contact the instructor as soon as possible.

Verbal and Written Communications

Communicating effectively in writing and verbally is critical to public health research and practice. Graduates of the Executive MPH program are expected to demonstrate these skills in written assignments and oral presentations. Requirements for acceptable performance in this course include the following:

Proofread: Clarity and accuracy are expected in submission of assignments throughout this course. Errors of grammar or spelling may result in a lower grade. Keep in mind that spell check and grammar check features of software programs such as Microsoft Word are not sufficient. Please read over all submissions carefully for content, spelling, punctuation, and grammar before turning them in.

Criteria for written assignments: The following criteria will be used in grading written assignments:

- Relevance to the issues
- Scope and thoroughness
- Logic and organization of thought
- Reference to the scientific literature in the areas related to the course
- Proper citations format (please use APA style – 6th edition)
- Integration of principles and theories described in class and in readings
- Overall writing style

- Correct grammar and spelling

Criteria for oral presentations: The following criteria will be used in grading oral presentations:

- Organization of material
- Relevance to the issues
- Scope and thoroughness
- Analysis and conclusions
- Presentation skills
- Adherence to time limits

Communicating with the Instructor

There are three primary modes of communication for this class – the "Course Q&A" and "Ask the Instructor" sections of the discussion board as well as electronic mail. To ensure that your questions are answered as promptly as possible, please follow the communications guidelines below:

- **Course Q&A:** This is a special forum on the discussion board that has been set up for **non-content** questions/answers about the course. This area will be monitored daily. **You are strongly encouraged to respond to your peers** if you know the answer or can provide guidance.
- **Ask the Instructor:** This forum is for content and assignment related questions for your peers and the instructor. The instructor will answer questions within 24-48 hours, so please do your best to anticipate questions by not waiting to complete assignments at the last minute.
- **Emory/Canvas email** to the instructor: Email should be used only for messages that are private in nature. Please allow 24-48 hours for response time.

There will be a weekly instructor office hour (day and time to be set pending student input during opening weekend). If students do not sign in to office hours within the first few minutes (or let the instructor know in advance of any delays) the session will be closed. Students are encouraged to schedule an appointment if it is not possible to attend these office hours.

Inclusive Learning Environment

As the instructor of this course I endeavor to provide an inclusive learning environment. However, if you experience barriers to learning in this course, do not hesitate to discuss them with me and the Office for Equity and Inclusion, (404) 727-9877.

Should content arise that is emotionally charged, it is important to remember that you are in a learning community composed of a wide range of beliefs and experiences, and every person has a right to express those beliefs/experiences within the discussion boundaries. No denigration or silencing of others will be allowed.

RSPH POLICIES

Accessibility and Accommodations

As the instructor of this course I endeavor to provide an inclusive learning environment. However, if you experience barriers to learning in this course, do not hesitate to discuss them with me and the Office of Accessibility Services (OAS). OAS works with students who have disabilities to provide reasonable accommodations. In order to receive consideration for reasonable accommodations, you must [register with the OAS](#). Once registered, you will be able to initiate a request for accommodations through **Accommodate**, a confidential, online platform. Upon approval of your accommodations by OAS, you may choose which classes you want the accommodations to apply to. Instructors of the classes you designate will receive your accommodation letter electronically through **Accommodate**, and you will be copied on the notification sent to the faculty. It is your responsibility to request accommodations. Please note that accommodations are not retroactive and that disability accommodations cannot be provided until your accommodation letter has been electronically signed by the faculty in the **Accommodate** system. I can only sign the document once you and I have discussed a plan to implement your accommodations in my course. Therefore, I encourage you to contact me as early as possible in the semester to discuss this plan. For additional information about accessibility and accommodations, please contact the Office of Accessibility Services at (404) 727-9877 or accessibility@emory.edu.

Honor Code

You are bound by Emory University's Student Honor and Conduct Code. RSPH requires that all material submitted by a student fulfilling his or her academic course of study must be the original work of the student. Violations of academic honor include any action by a student indicating dishonesty or a lack of integrity in academic ethics. *Academic dishonesty refers to cheating, plagiarizing, assisting other students without authorization, lying, tampering, or stealing in performing any academic work, and will not be tolerated under any circumstances.*

The [RSPH Honor Code](#) states: "Plagiarism is the act of presenting as one's own work the expression, words, or ideas of another person whether published or unpublished (including the work of another student). A writer's work should be regarded as his/her own property."

COURSE CALENDAR AND OUTLINE

Semester Week	Date	Topics	SAS Terms	Lecture/Demo	Quizzes	Assignment
Opening Weekend	Fri 1/8, 4:00-7:00 pm Sun 1/10, 8:30-11:30 am	Administrative Intro to SAS SAS dataset basics SAS explorer applications SAS customizations SAS resources, best practices	LIBNAME EXPLORER VIEWTABLE ENHANCED EDITOR	In Class <i>Module 1 Video (optional)</i>		
Week 1	1/11-1/17	SAS dataset creation Concatenating and merging datasets	INPUT INFILE DATALINES PROC IMPORT PROC CONTENTS PROC PRINT PROC FREQ PROC UNIVARIATE PROC SORT SET MERGE	Module 2 Video	Quiz 1 due	Lab 1 due
Week 2	1/19 (Tues.)-1/24 (MLK Holiday, 1/18)	Sub-setting of SAS datasets Data cleaning Creating new variables	MERGE DO/END IN STATEMENTS OBS TITLE KEEP DROP IF/THEN WHERE CROSSTABS	Module 3 Video	Quiz 2 due	Lab 2 due
Week 3	1/25-1/31	Categorical data Performing arithmetic operations SAS dates and date functions	GT/LT/LT/GE PCTLPTS OUTPUT + - * / ** INT MDY YEAR, DATE() IF/THEN/ELSE	Module 4 Video	Quiz 3 due	Lab 3 due

Semester Week	Date	Topics	SAS Terms	Lecture/Demo	Quizzes	Assignment	
Week 4	2/1-2/7	More on IF/THEN/ELSE PUT and INPUT to convert values	IF/THEN/ELSE PUT INPUT	Module 5 Video	Quiz 4 due	Lab 4 due	
Week 5	2/8-2/14	Re-naming, labeling, and formatting variables Useful procedures Output delivery system basics	RENAME LABEL PROC FORMAT ODS RTF ODS PDF PROC COMPARE PROC TRANSPOSE	Module 6 Video	Quiz 5 due	Lab 5 due	
Week 6	2/15-2/21 (Mid-Semester Course Evals)	MIDTERM					
Week 7	2/22-2/28	Fun with functions	COMPRESS SUBSTR UPCASE LOWCASE PROPCASE SCAN FIND TRANWRD MDY MONTH WEEKDAY DAY YEAR YRDIF SMALLEST	LARGEST TRIMN STRIP CEIL FLOOR INT ROUND ZIPFIPS ZIPNAME ZIPNAMEL ZIPSTATE COUNTC CATS ARRAY	Module 7 Video	Quiz 6 due	Lab 6 due
Week 8	3/1-3/7	Using arrays Naming of variables in datasets	ARRAY	Module 8 Video	Quiz 7 due	Lab 7 due	

Semester Week	Date	Topics	SAS Terms	Lecture/Demo	Quizzes	Assignment
Week 9	3/8-3/14 (Reading Wk: 3/8-3/12)	READING WEEK				
Week 10	3/15-3/21	Using datasets with multiple observations per subject	PROC SUMMARY PROC MEANS SET BY FIRST./LAST. CLASS RETAIN NODUPKEY	Module 9 Video	Quiz 8 due	Lab 8 due Article for presentation due
Week 11	3/22-3/28	More on reporting/output Exporting data	PROC REPORT PROC TABULATE EMAIL PROC EXPORT	Module 10 Video	Quiz 9 due	Lab 9 due
Week 12	3/29-4/4	Intro to macros	& % %LET	Module 11 Video	Quiz 10 due	Lab 10 due
Week 13 – Short Week	4/5-4/8	Intro to graphs/charts/maps Extra topic(s) TBD	PROC SGPLOT PROC SGPANEL PROC GMAP	Module 12 Video	Quiz 11 due	Lab 11 due
Closing Weekend	Fri 4/9, 4:00-7:00 pm Sun 4/11, 8:30-11:30 am	Article Presentation (Fri) Review session (Fri) Final exam (Sun)		In class		

Note: Modules will open on the Friday before the week begins at 12:00 pm EST. All assignments are due on Sunday of the weekly module at 11:59 pm ET.